Zeqian Yu

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EDUCATION

MRes in Economic Analysis — Madrid, Spain
Universidad Carlos III de Madrid — 2026 – Present

Master in Economics: Empirical Applications and Policies — Bilbao, Spain
University of the Basque Country (UPV/EHU) — 2026

Master in Finance — Barcelona, Spain
EADA Business School Barcelona — 2018

Bachelor of Economics Sciences in Finance — Xi'an, China
Xi'an International Studies University — 2017

WORKING PAPERS

High-Speed Rail Expansion and Electric Vehicle Adoption in Spain — Revise and resubmit
SERIEs: Journal of the Spanish Economic Association. Master's thesis (10/10, with Distinction), UPV/EHU; supervised by Prof. Javier Gardeazabal.

- Uses the staggered expansion of Spain's HSR network as a quasi-experiment to estimate the causal effect of HSR connectivity on municipal-level EV adoption, implementing staggered difference-in-differences (DID) and Callaway and Sant'Anna DID (CSDID) estimators on a Mahalanobis-matched sample of comparable municipalities, with a supporting least-cost instrumental variable design.
- Finds that HSR connection raises the battery electric vehicle (BEV) share of new registrations, robust to local charging infrastructure and purchase subsidies, suggesting that HSR access alleviates consumers' range anxiety by providing a substitute for long-distance travel, identifying the transport network as a vital complement to the electric transition.

SELECTED COURSE PROJECTS

Electricity Market Dynamics and Pricing

- Measured concentration in Spain's intraday electricity market (2021–2025) using a range of market-concentration indices and tracked how it shifted over time.
- Reconstructed supply and demand curves from day-ahead hourly bidding data (2020) to estimate equilibrium prices, comparing alternative functional-form specifications.
- Analyzed daily electricity prices for Ireland (2015–2025) with time-series models to assess price persistence, seasonality, and volatility.

Web Scraping and Sentiment Analysis of Research Publications

- Built an automated pipeline to scrape a researcher's full publication record from Google Scholar, assembling a structured dataset of titles, authors, publication years, citations, and journals.
- Applied lexicon-based sentiment analysis to trace how the tone of publication titles evolved over time, and packaged the results in an interactive web application.

Student Achievement Prediction Using Machine Learning

- Predicted student math achievement from international PISA assessment microdata for Hong Kong, cleaning the data and preparing a broad set of demographic, socioeconomic, attitudinal, and school-climate predictors.
- Built, trained, and tuned a full machine-learning workflow spanning Ridge, LASSO, Elastic Net, Decision Tree, Bagging, Random Forest, and Gradient Boosting, then compared their accuracy to select the strongest predictive model.
- Used dimensionality reduction and clustering to profile students by socioeconomic background and learning attitudes.

WORKSHOPS, SEMINARS & CONFERENCES

Data Science Summer School Berlin, Germany (Remote)
Hertie School Data Science Lab 2025
- Core courses in R and Python programming, applied calculus, linear algebra, and statistics and probability, with advanced applications in AI agents and experimental design.

Computational and Analytic Techniques in the Humanities and Social Sciences Nicosia, Cyprus (Remote)
University of Cyprus 2025
- Introductory training in GIS and remote sensing for social-science research.

Series of Academic and Professional Courses Self-paced (Online)
Santander Open Academy 2024
- Completed courses including Introduction to Data Science, Generative AI, Digital Transformation, and Internet of Things.

Academic Writing for the Health and Social Sciences Vic, Spain (Remote)
University of Vic – University of Central Catalonia 2023
- Structuring research articles and building academic writing skills.

Presentations at Congresses, Conferences and Workshops Vic, Spain (Remote)
University of Vic – University of Central Catalonia 2023
- Practical tools for delivering research presentations in English.

Graphic Design for Presentations and Scientific Communication Vic, Spain (Remote)
University of Vic – University of Central Catalonia 2023
- Building skills to design clearer, more effective visuals for scientific communication.

EURAXESS Science Slam China Science Communication Training Beijing, China (Remote)
EURAXESS 2023
- Online science-communication training in scriptwriting, research pitching, and short science videos; awarded the top-level Diamond certificate.

Public Speaking and Investor Pitching Toruń, Poland
Nicolaus Copernicus University 2022
- Delivering clear presentations and pitching ideas effectively.

Data Science Summer School Berlin, Germany (Remote)
Hertie School Data Science Lab 2022
- Advanced machine-learning and deep-learning applications in computer vision and natural-language processing.

EURAXESS Science Slam China Science Communication Training Beijing, China (Remote)
EURAXESS 2021
- Online science-communication training in public speaking, creative pitching, and audience engagement; awarded the top-level Diamond certificate.

RELEVANT PROFESSIONAL EXPERIENCE

E-House (China) Holdings Limited, CRIC Shenzhen, China
Consultant, Data Service Department Mar. 2019 – Aug. 2022
- Conducted market research and financial analysis to identify trends and deliver actionable insights for clients in the real-estate industry.
- Led the creation of monthly market reports, from initial analysis to refined, client-ready outputs, and coauthored two industry white papers on the Greater Bay Area.
- Supervised and mentored a team to deliver high-quality, data-driven research reports.

HONORS & AWARDS

Full Tuition Waiver, MRes in Economic Analysis 2026
Universidad Carlos III de Madrid

OTHER INFORMATION

Languages: Chinese (native); English (C1)
Technical Skills: R; Stata; ArcGIS; Beginner in MATLAB; LaTeX
Certificates: CFA Level I

Last updated: 3 November 2026